

Urbanization and CO2 Emissions in Developing Countries

MATH318 | Statistical Learning
Project



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Research Question

How does urbanization affect CO2 emissions in developing countries, and is there evidence of an environmental turning point?



Introduction

- Urbanization can improve a countries living conditions, however, it may lead to increase CO2 emissions
- We go through pooled OLS models, as well as country and time fixed effects.
- Using panel data methods, we will distinguish between cross-country relations and within country dynamics



Data & EDA

Data Description

- Panel data = Cross sectional data + Time series
- There are 28 years of data for 36 countries
- All variables in the data set are varying regressors (except 1)



Data & EDA

Data Description

Variables:

country	—	Country name
co2.pc	—	CO2 emissions per capita (<i>target</i>)
pop	—	Total population
en.pc	—	Energy consumption per capita
gdp.pc	—	GDP per capita
lagco2.pc	—	Lagged CO2 emissions per capita
co2.pc_gwt	—	CO2 emissions per capita growth
pop_gwt	—	Total population growth
gdp.pc_gwt	—	GDP per capita growth
urb	—	Urbanization % (<i>main regressor</i>)
ci	—	Carbon index
edu	—	Education index
id	—	Country ID



Data & EDA

Exploratory Data Analysis

01 None of the entries are missing

02 All variable are float except for country

03 None of the countries have typos

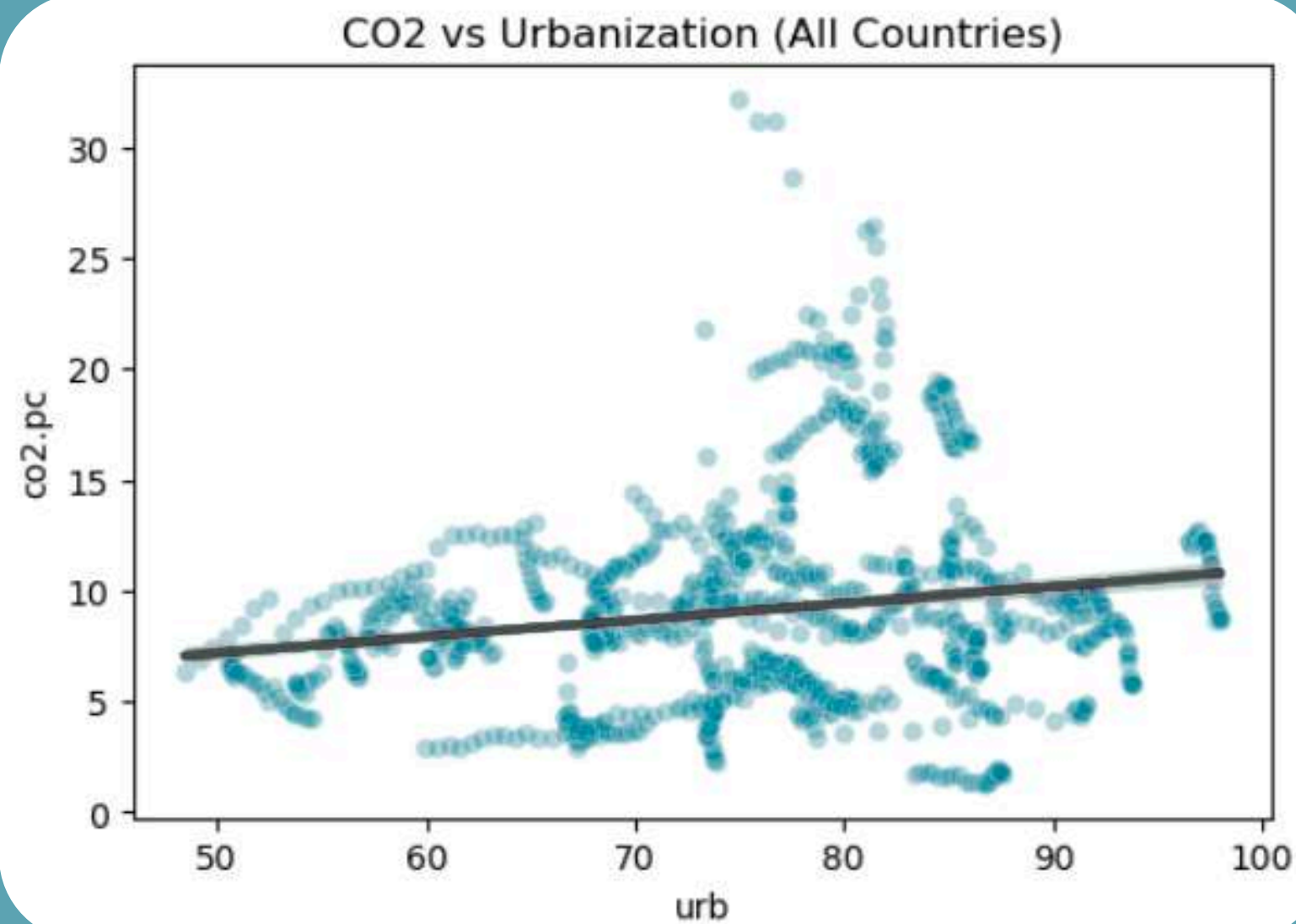
04 There are 28 observations per country for every year

05 We will add a variable for the years 1975 to 2002

06 The dataset is clean

Data & EDA

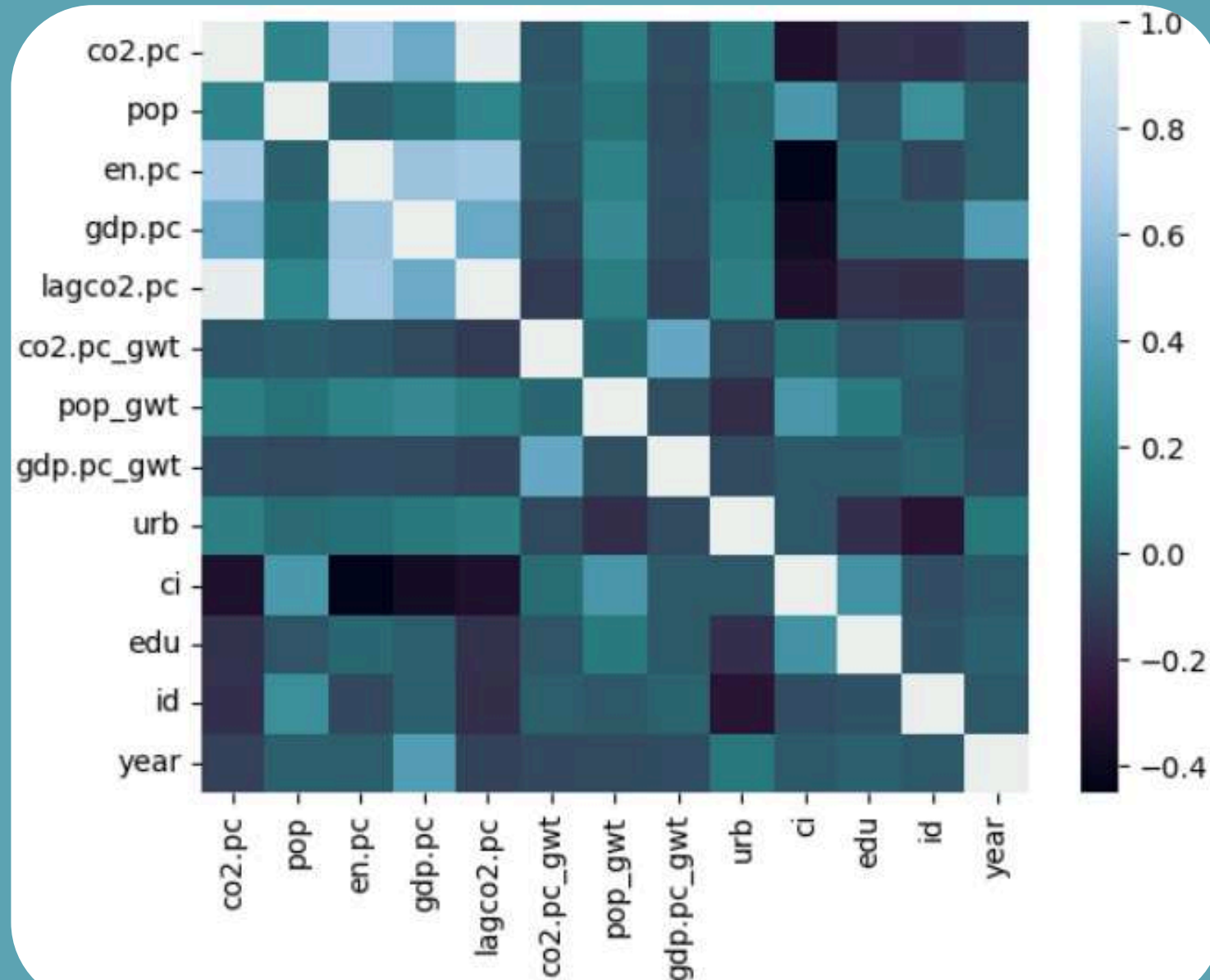
Scatter Plot



- This pooled plot suggests a positive relationship across countries
- There seems to be some hidden relationship
- Possible depending on the country

Data & EDA

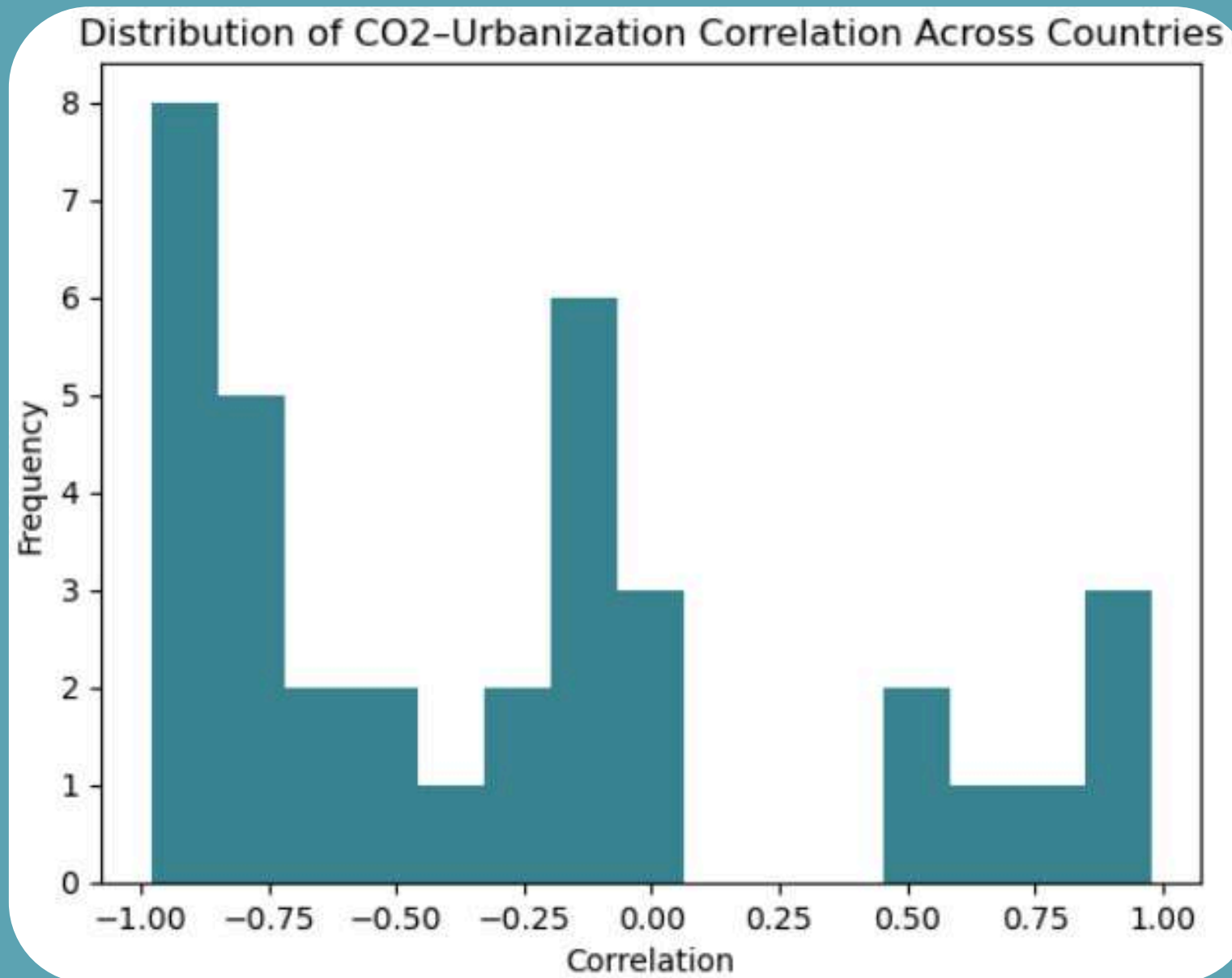
Correlation Heatmap



- CO2 emissions are positively correlated to energy consumption and GDP
- Relationship with urbanization looks weak
- The heatmap masks underlying trends for each country

Data & EDA

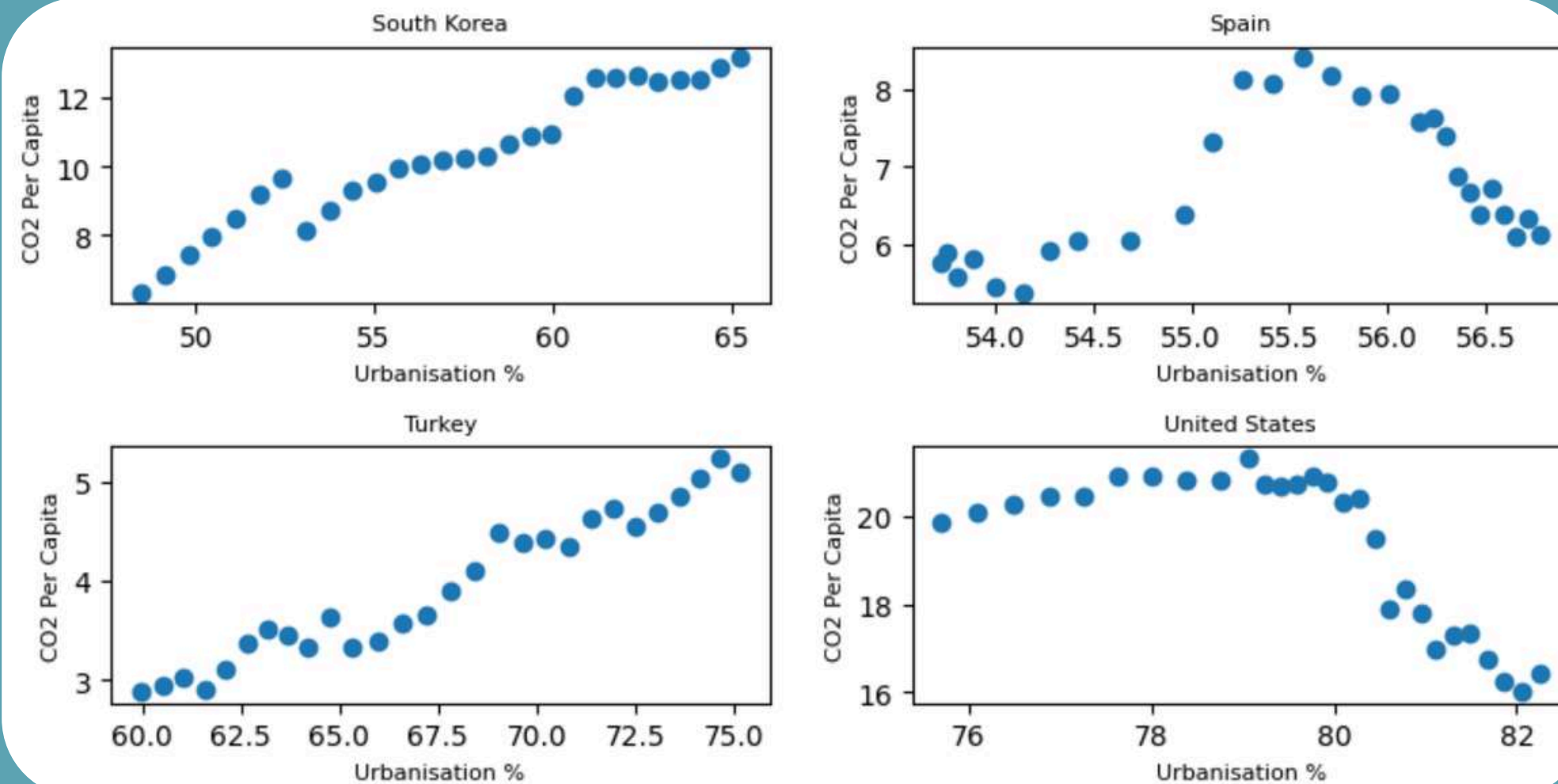
Histogram of Correlation



- Most countries show strong negative correlation between CO2 emissions and Urbanization
- Effect of urbanization is not consistent across countries

Data & EDA

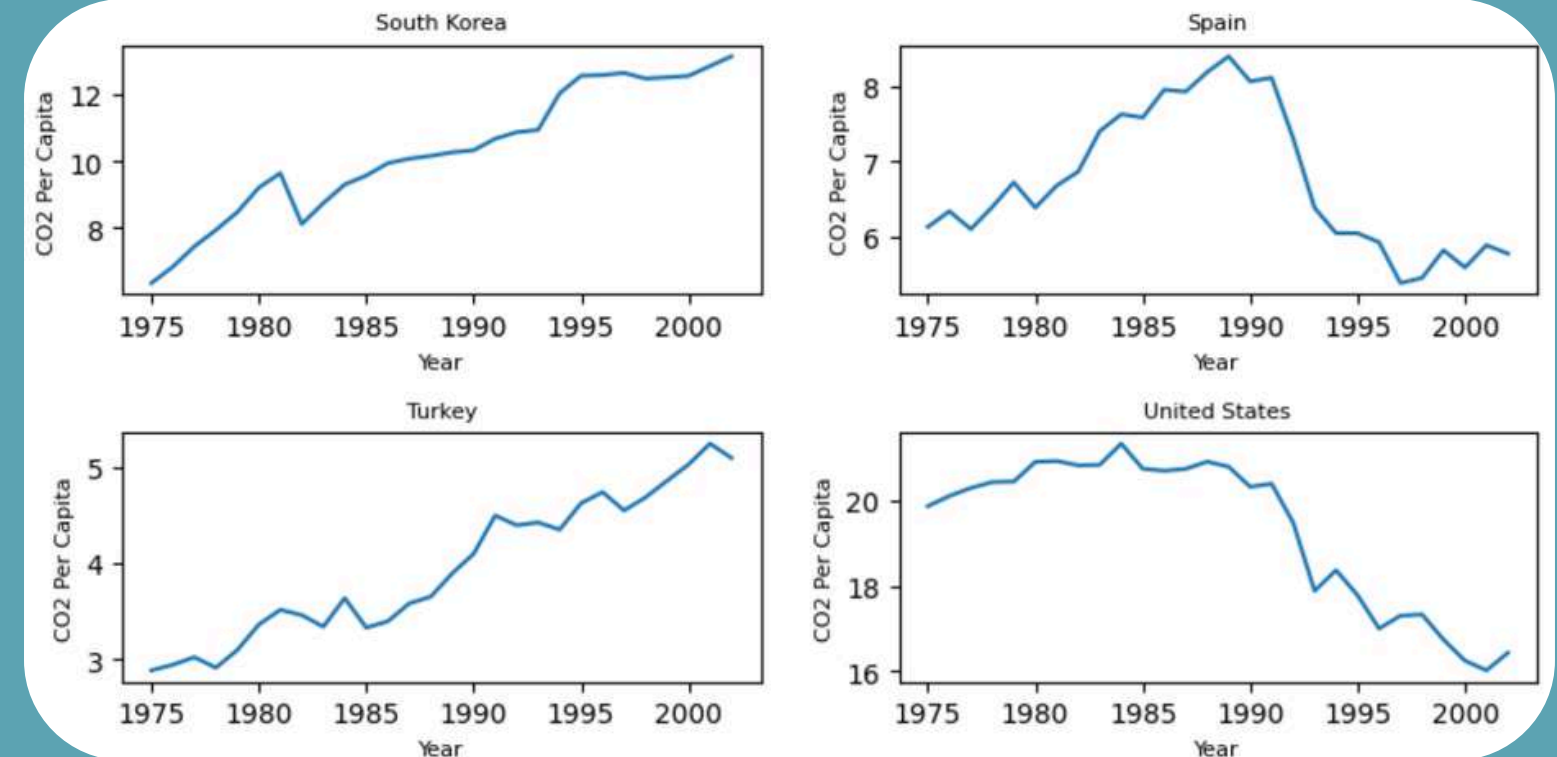
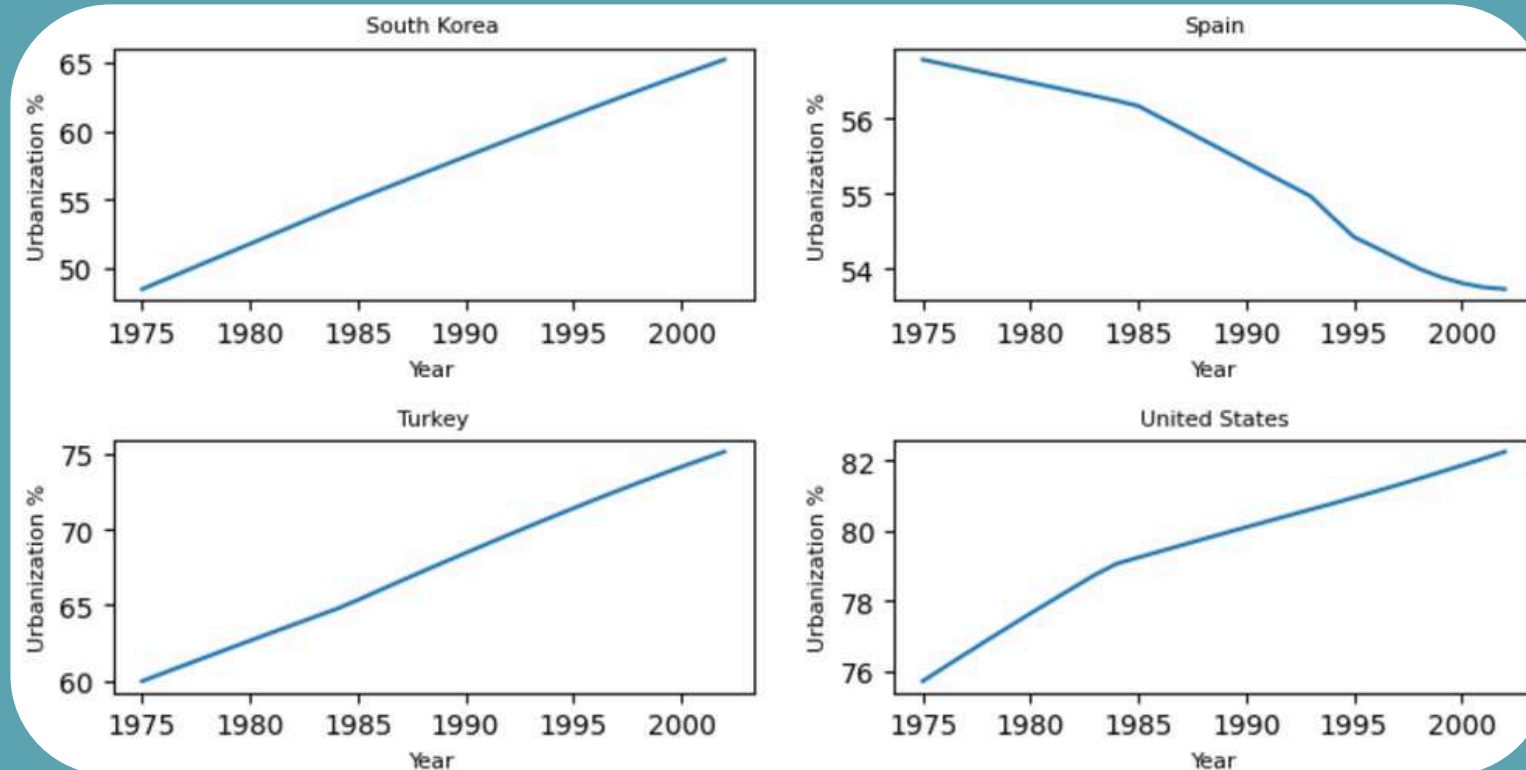
Cross-country scatter plot



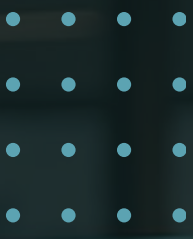
- Splitting the countries allowed us to see specific trends
- There is evidence of linear relationships, as well as quadratic relationships

Data & EDA

Time-Series plots across countries



- CO2 emissions and urbanization changes differently over time for each country
- Presenting panel data in this way lets us see relationships more clearly



Methodology

Models



Linear Pooled OLS

The first suggested baseline model was a linear pooled OLS.



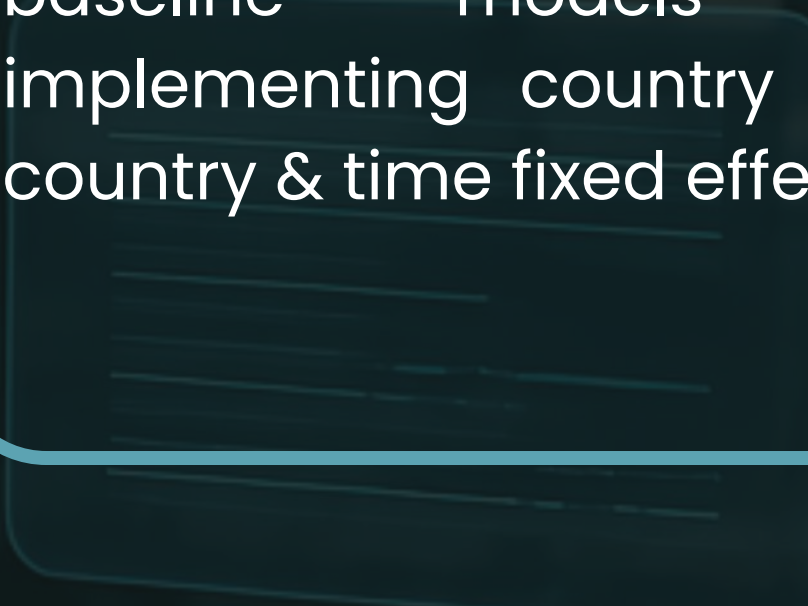
Quadratic Pooled OLS

Next we test potential non-linearity by including a quadratic urbanization term.



Fixed Effects

We extended beyond the baseline models by implementing country and country & time fixed effects.



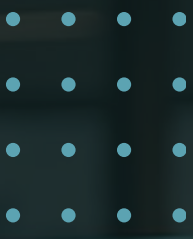
Methodology

Models

Linear Pooled OLS

The first suggested baseline model was a linear pooled OLS. We hold GDP and energy constant to interpret a linear relationship between CO2 emissions and Urbanization.

$$CO_{2,it} = \beta_0 + \beta_1 Urban_{it} + \beta_2 GDP_{it} + \beta_3 Energy_{it} + u_{it}$$



Methodology

Models

Linear Pooled OLS

The first suggested baseline model was a linear pooled OLS.

Quadratic Pooled OLS

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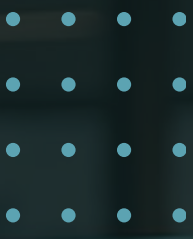
Methodology

Models

Quadratic Pooled OLS

Next we test potential non-linearity by including a quadratic urbanization term. This allows us to test for a potential turning point in the data. We also keep GDP and energy constant.

$$CO_{2,it} = \beta_0 + \beta_1 Urban_{it} + \beta_2 Urban_{it}^2 + \beta_3 GDP_{it} + \beta_4 Energy_{it} + u_{it}$$



Methodology

Models

Linear Pooled OLS

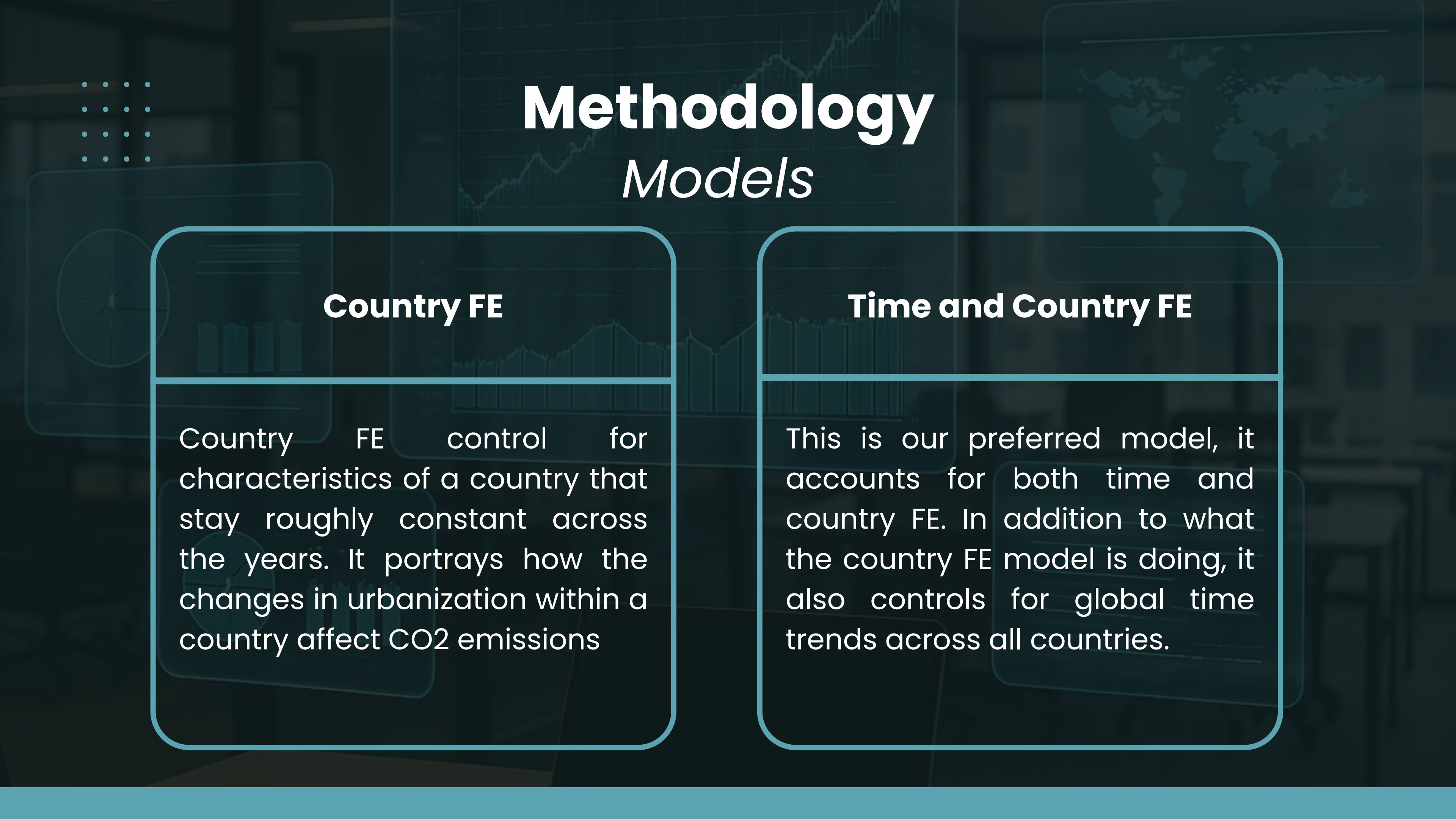
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Quadratic Pooled OLS

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Methodology

Models

Country FE

Country FE control for characteristics of a country that stay roughly constant across the years. It portrays how the changes in urbanization within a country affect CO2 emissions

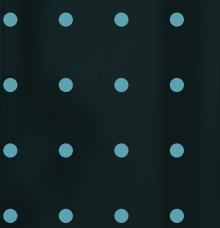
Time and Country FE

This is our preferred model, it accounts for both time and country FE. In addition to what the country FE model is doing, it also controls for global time trends across all countries.

Methodology

Diagnostics and Inference

Residual diagnostics revealed heteroskedasticity and potential correlation of errors within countries over time, violating standard OLS assumptions.



Linear Pooled OLS

- **Urbanisation** — $\beta = 0.0524$, $p = 0.0001$
 - Statistically significant
 - Positively associated with CO₂ emissions
- **Energy Consumption** — $\beta = 0.0001$, $p < 0.0001$
 - Strongly statistically significant
 - Positively associated with CO₂ emissions
- **GDP Per Capita** — $p = 0.00554$
 - Only marginally statistically significant
 - Suggests *weak, indirect* effect on CO₂
- **Predictive Performance**
 - MSE = 12.55

Quadratic Pooled OLS

- **Urbanisation** — $\beta = 0.4667$, $p = 0.0009$
 - Statistically significant
 - Positively associated with CO₂ emissions
- **Urbanisation²** — $\beta = -0.0028$, $p = 0.0031$
 - Statistically significant
 - Negatively associated with CO₂ emissions
- **Turning Point**
 - Inverted U-shape at 82.3% urbanisation
- **Predictive Performance**
 - MSE = 12.40

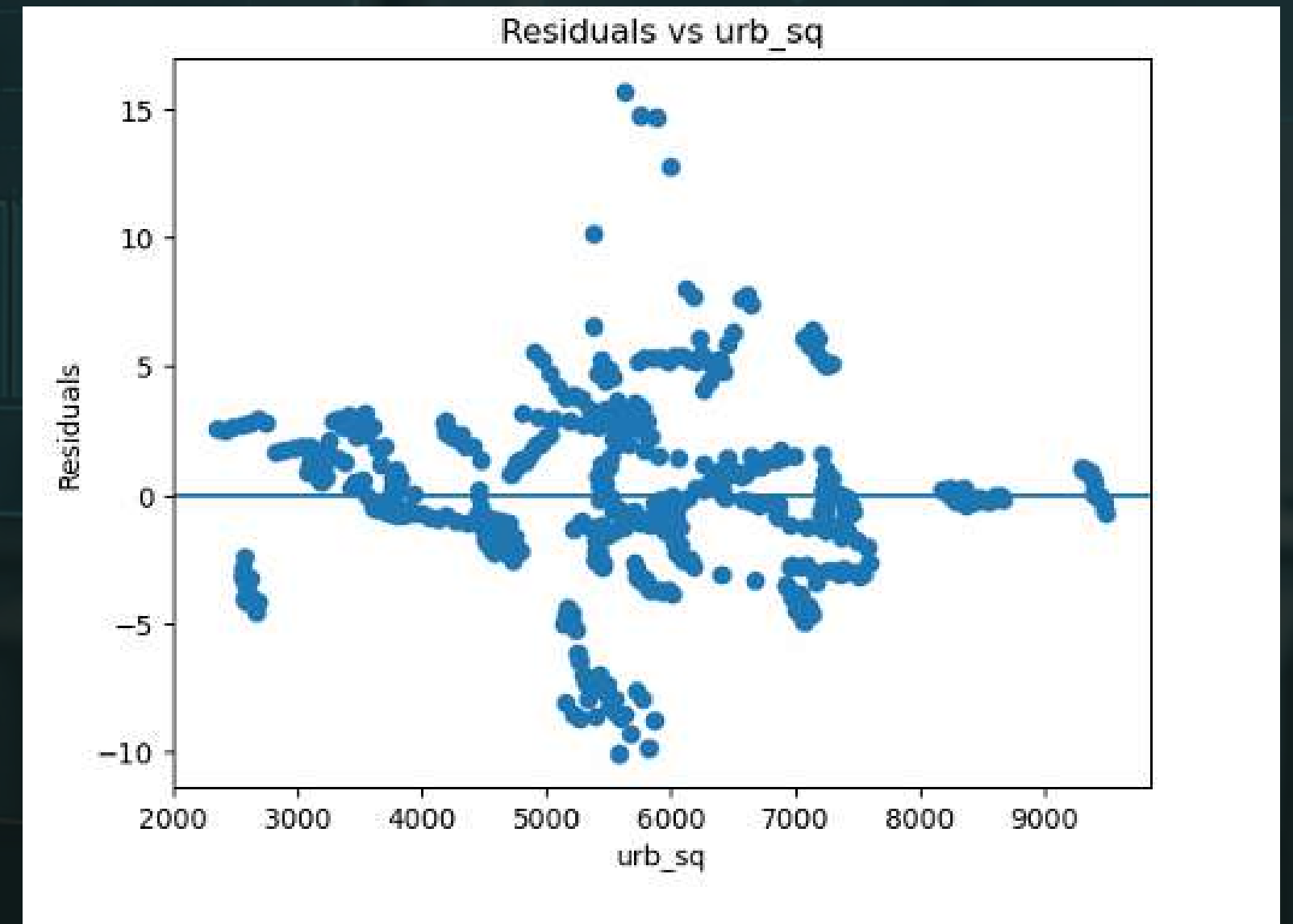
Variable Selection & Diagnostics

Variable Selection – Best Subset Selection

- Alternative models perform slightly better
- *Baseline specification retained* for theoretical relevance and parsimony

Residuals & Multicollinearity Checks

- Residuals vary with predictors; heteroskedasticity
→ *use clustered standard errors*
- No multicollinearity (low VIFs)



Country Fixed Effects

- Controls for unobserved, cross-country differences
 - Allows identification of *within-country relationships*
 - Key results:
 - Energy remains statistically significant
 - GDP not even marginally significant
 - Adding a quadratic term yields *no significant evidence of nonlinearity*; implied turning point lies outside observed range.
- **Urbanisation coefficient is *negative***

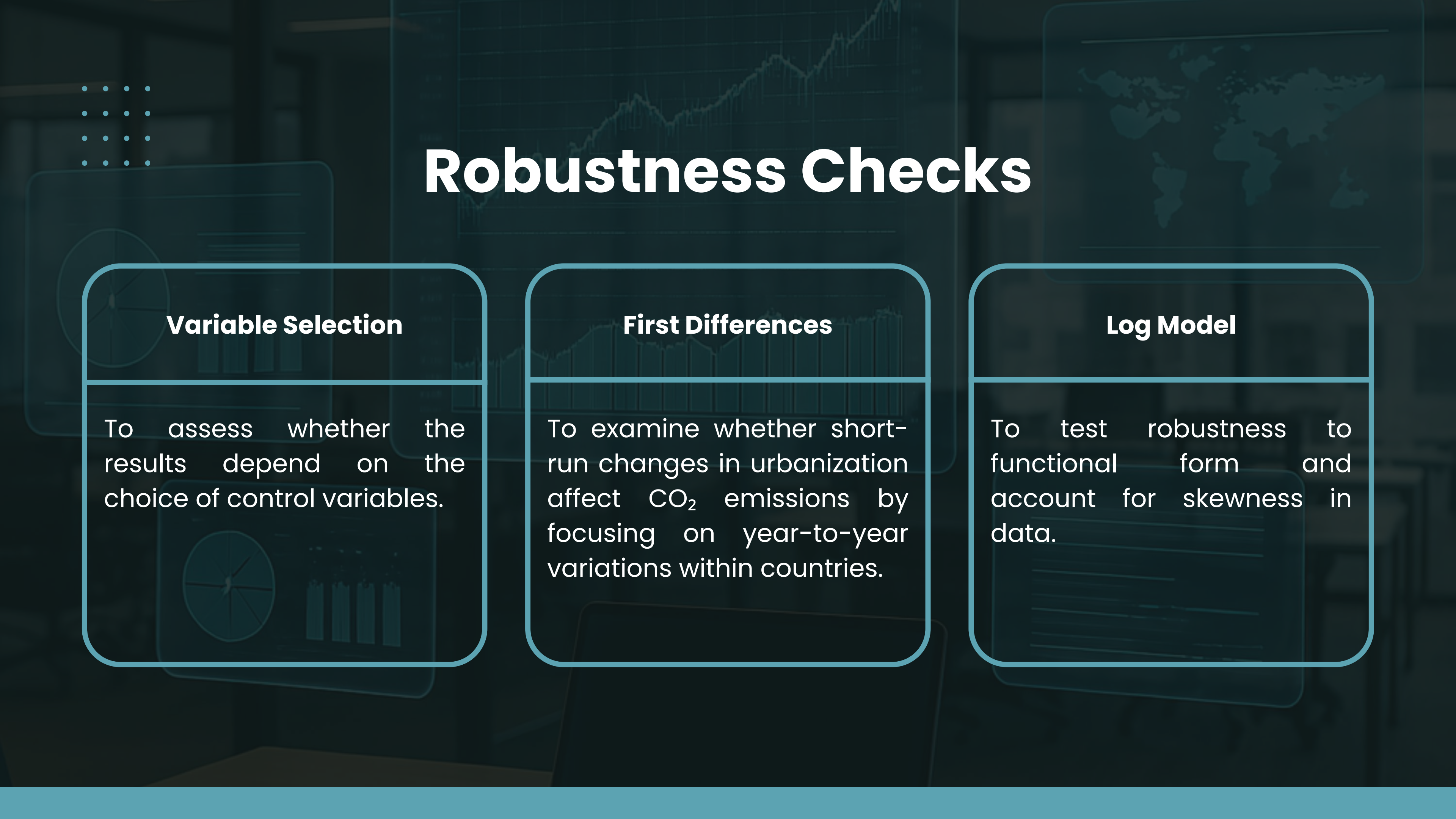
Country & Time Fixed Effects

- Controls for *time and cross-country differences*

- Key results:

- Energy remains statistically significant
- GDP remains statistically insignificant

- **Urbanisation is *not statistically significant***



Robustness Checks

Variable Selection

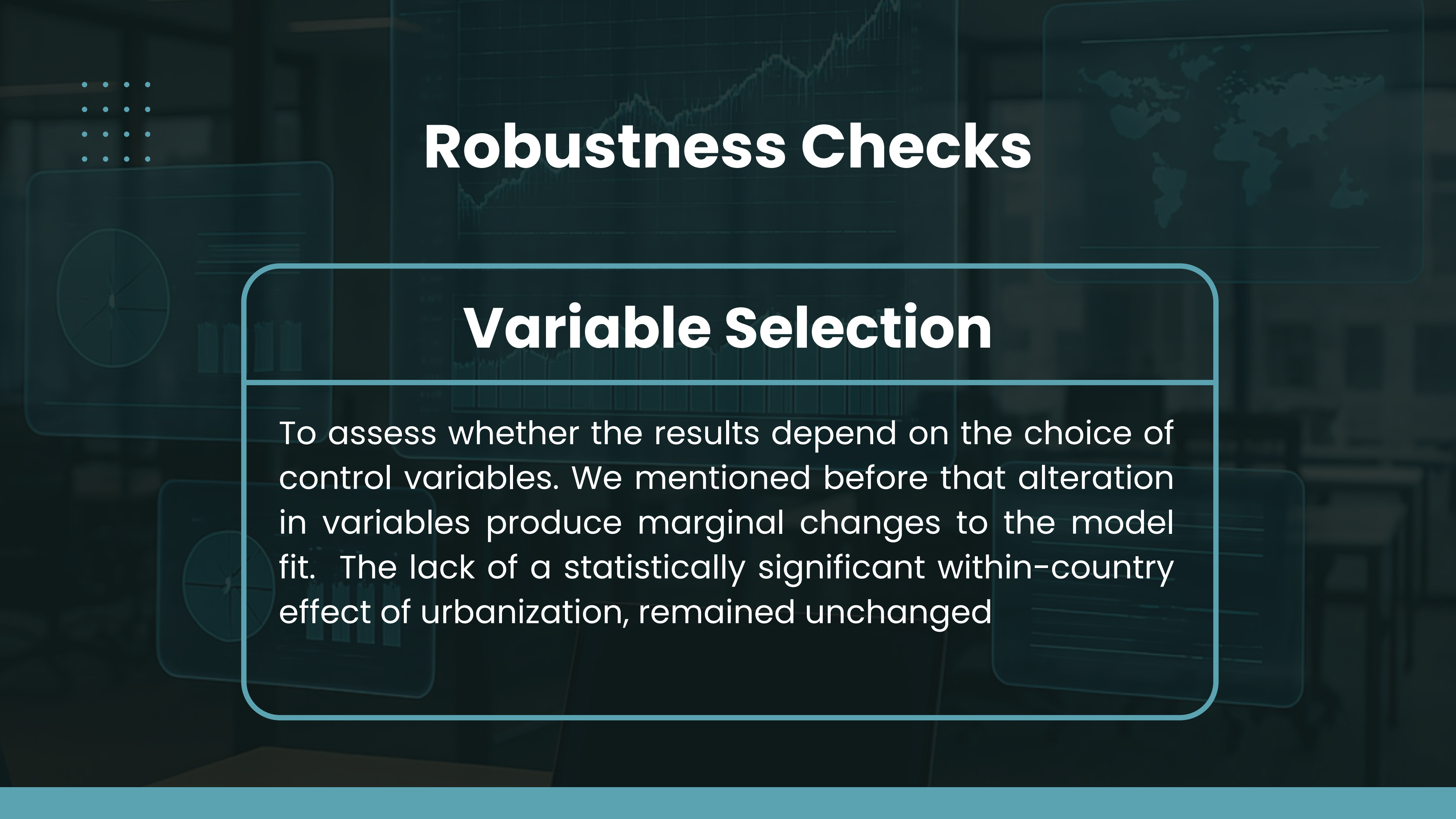
To assess whether the results depend on the choice of control variables.

First Differences

To examine whether short-run changes in urbanization affect CO₂ emissions by focusing on year-to-year variations within countries.

Log Model

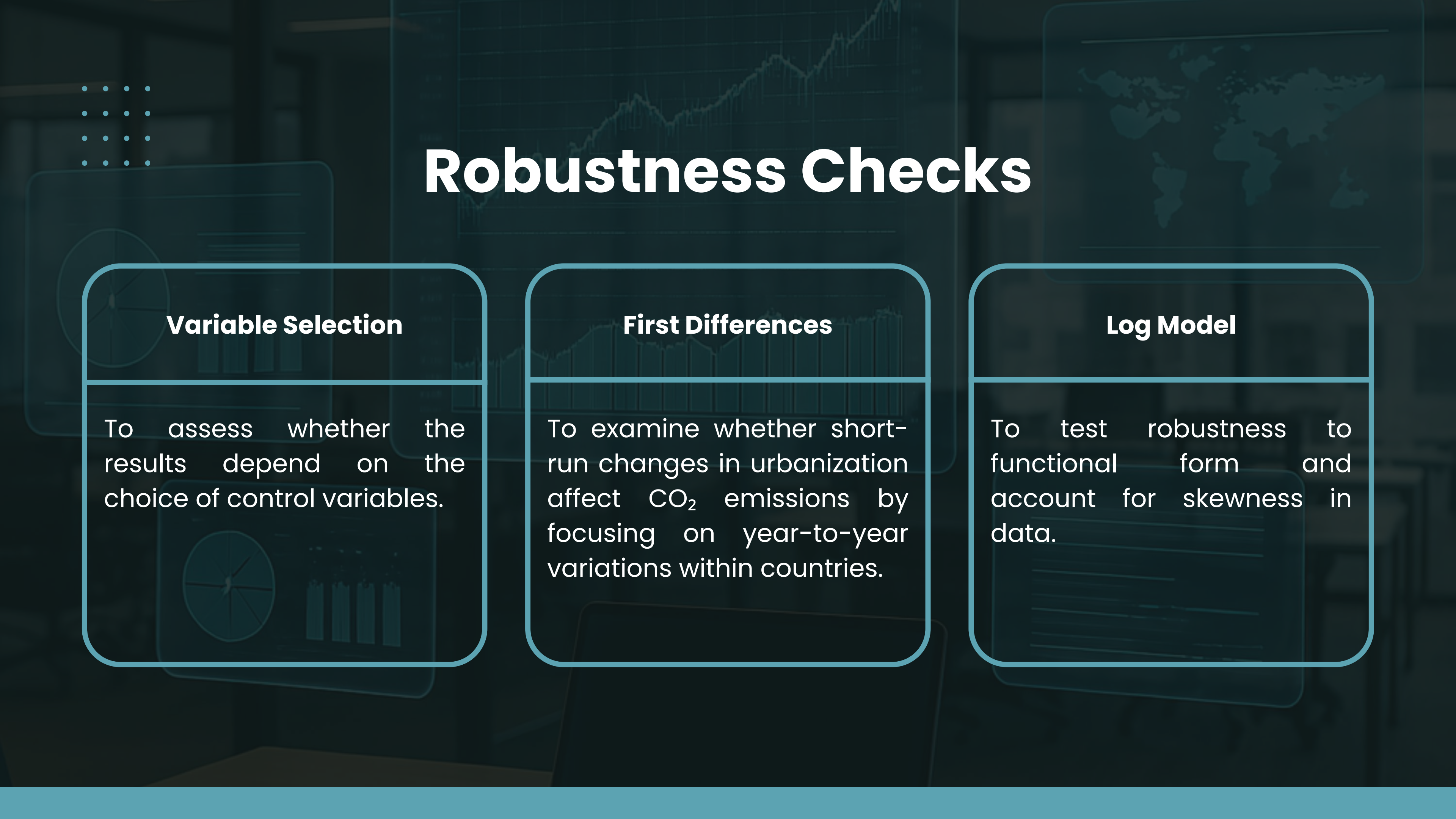
To test robustness to functional form and account for skewness in data.



Robustness Checks

Variable Selection

To assess whether the results depend on the choice of control variables. We mentioned before that alteration in variables produce marginal changes to the model fit. The lack of a statistically significant within-country effect of urbanization, remained unchanged



Robustness Checks

Variable Selection

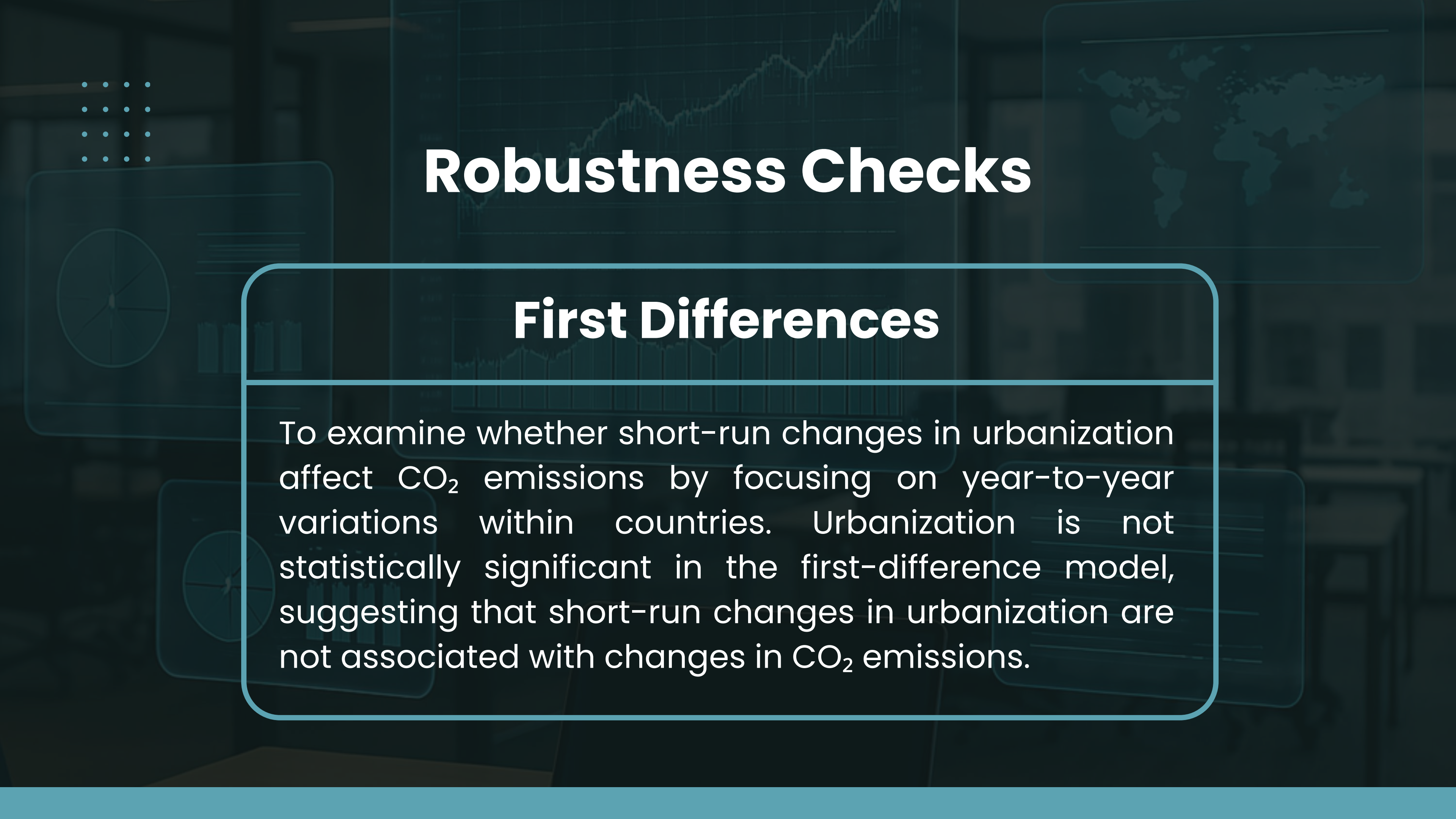
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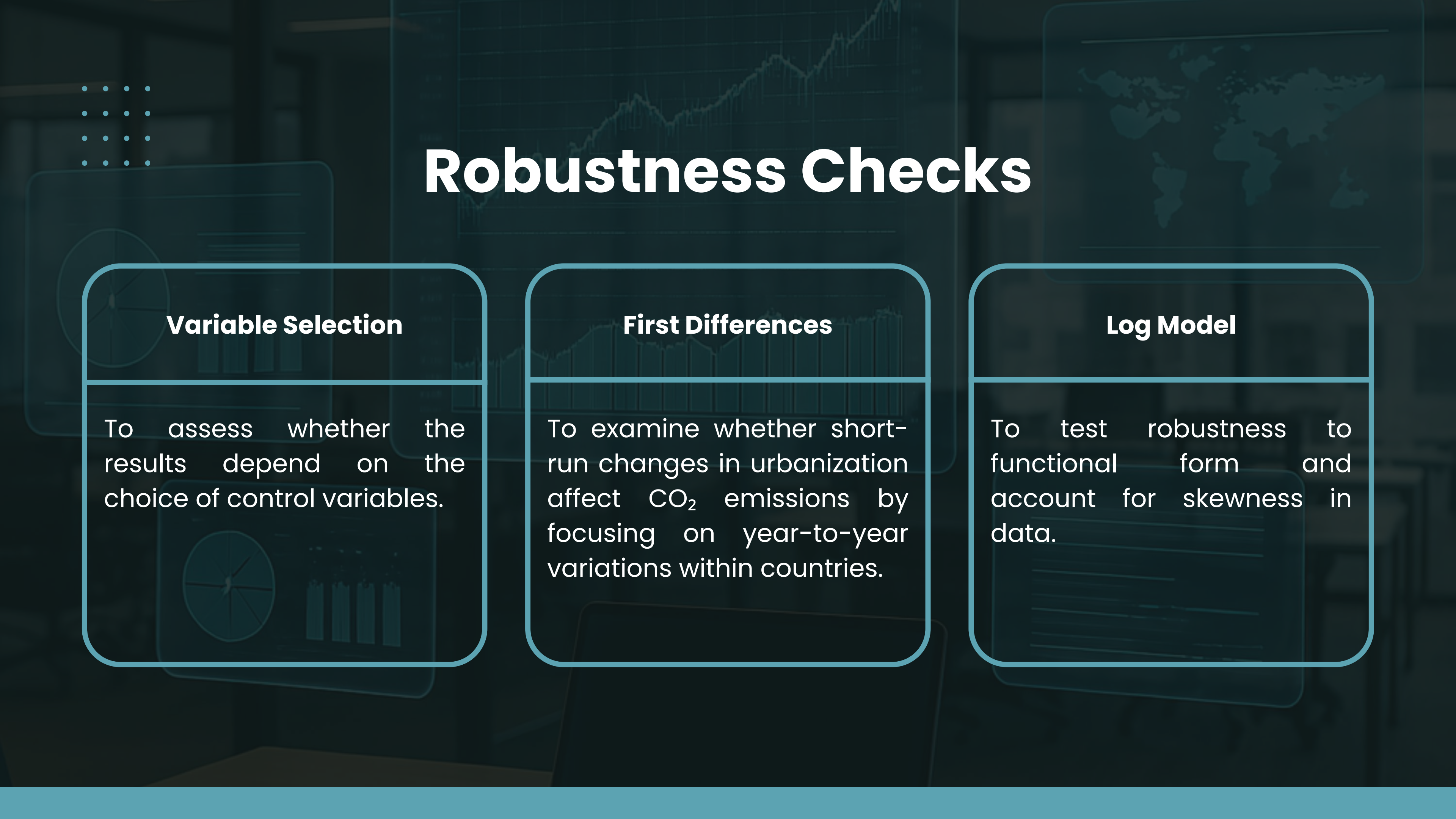
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Robustness Checks

First Differences

To examine whether short-run changes in urbanization affect CO₂ emissions by focusing on year-to-year variations within countries. Urbanization is not statistically significant in the first-difference model, suggesting that short-run changes in urbanization are not associated with changes in CO₂ emissions.



Robustness Checks

Variable Selection

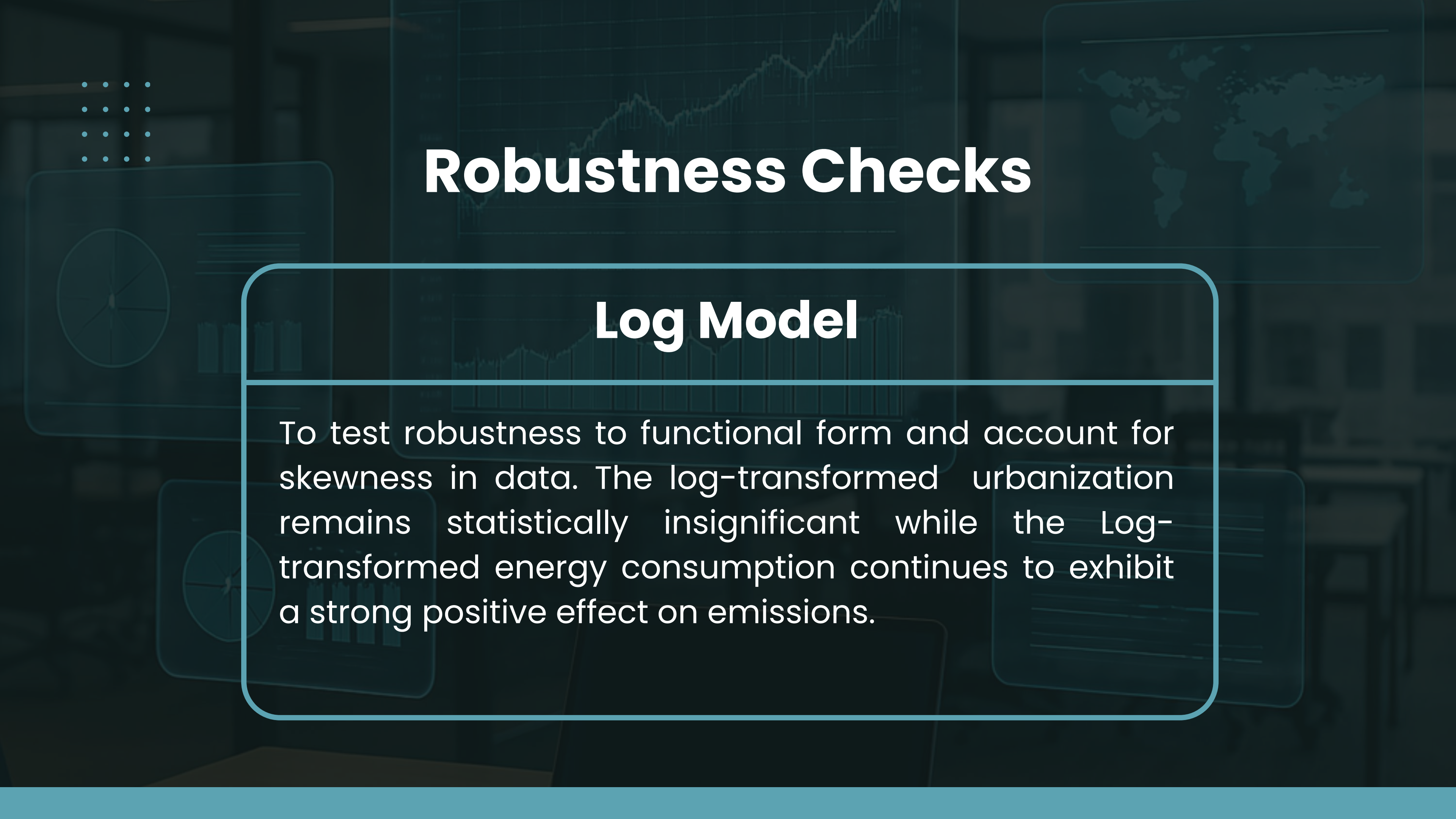
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First Differences

To examine whether short-run changes in urbanization affect CO₂ emissions by focusing on year-to-year variations within countries.

Log Model

To test robustness to functional form and account for skewness in data.



Robustness Checks

Log Model

To test robustness to functional form and account for skewness in data. The log-transformed urbanization remains statistically insignificant while the Log-transformed energy consumption continues to exhibit a strong positive effect on emissions.

Robustness Checks

Variable Selection

To assess whether the results depend on the choice of control variables.

First Differences

To examine whether short-run changes in urbanization affect CO₂ emissions by focusing on year-to-year variations within countries.

Log Model

To test robustness to functional form and account for skewness in data.

Overall, the findings remain consistent across specifications

Conclusion

Discussion

- It is important to distinguish between cross-country and within-country effects
- Pooled models and fixed effects models gave different results
- Urbanization by itself is not the direct cause of higher CO₂ emissions

Conclusion

Heterogeneity

- Heterogeneity analysis based on GDP and population shows that the estimated relationship between urbanization and CO₂ emissions does not differ across the studied country characteristics

Conclusion

Interpretation and Policy

- Across all model specifications, energy consumption remained a strong predictor of CO2 emissions
- Urbanization indirectly influences CO2 emissions through other factors
- Plausible mechanisms: increased energy demand and infrastructure development

Conclusion

Interpretation and Policy

- Efforts to reduce CO2 emissions should focus more on developing ways to improve energy efficiency
- Urbanizations that are well-managed could contribute to the increase of more efficient energy usage.



Thank you